

JOSHUA DALAL

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EXPERIENCE

Pratt & Whitney · Engine Dynamics & Loads Intern

May 2026 – Present · East Hartford, CT

- Building data-driven models that correlate aircraft structural characteristics with engine loads
- Improving reproducibility of NASTRAN analyses for engine loading due to fan blade-off events
- Characterizing squeeze-film damper performance experimentally to quantify how operating conditions drive vibration response and dynamic loads in rotating machinery
- Supporting whole-engine structural response evaluations that generate engine loading conditions

Hanwha Aerospace USA · Manufacturing Engineering Intern

Jun 2025 – Aug 2025

- Owned process flow design of the F135 rotor for the Engine Core Upgrade; modeled part operations and drafted operation sheets for 27 manufacturing, inspection, and cleaning operations
- Generated PFMEA database for a Hanwha Korea part, restructured for assembly-level operations
- Developed and documented fixture and tooling designs, driving a rate-increase clawback of \$100k
- Qualified abrasive flow machining for a new part, improving surface finish and reducing cost 35%

Albacore Inc. · Propulsion & Systems Engineer

Jan 2025 – Jun 2025

- Developed an Unmanned Underwater Vehicle (UUV) for the US Navy, with a novel powertrain achieving 10x the range of existing models at 4x lower cost
- Designed CAD parts and assemblies and managed vehicle-level mechanical system integration
- Prototyped key subsystems and ran tests to validate design feasibility and performance
- Supported US Navy SBIR submissions; credited with the filing of a provisional patent

GRASP Robotics Lab, UPenn · Research Assistant

May 2024 – Nov 2024

- Designed a highly articulated robotic arm driven by one base-mounted motor, cutting weight by eliminating motors at each joint
- Engineered a u-joint and bevel-gear transmission to minimize backlash and optimize joint locking
- Designed and 3D-printed prototypes in Onshape, refining parts for durability and precision
- Prototyped a mechanical eyeball for an Air Force eye-tracking project: 0.3° accuracy at 600°/s

LEADERSHIP & PROJECTS

Project Wavefront · Co-Founder & Propulsion Engineer

Apr 2025 – Present

- Leading design of an entirely student-built rotating detonation rocket engine (RDRE): a monolithic, 3D-printed GOX/GCH₄ engine targeting 2 kN thrust, 255 s Isp, and 21 bar chamber pressure
- Designed water-cooled architecture rejecting 0.95 MW of heat for 30+ second runtime, with integrated DDT ignition and triplet impinging injectors supporting 13.5 kHz wave operation
- Simulating RDRE injector mixing and chamber flow with ANSYS Fluent CFD and Python tools
- Operating a 501(c)(3) nonprofit: fundraising, research outreach, and industry partnerships

Penn Jet Propulsion · Propulsion Team Lead

Aug 2023 – Present

- Leading a hydrogen turbofan research project on sustainable combustion through micromixing; overseeing combustor geometry, fuel-air mixing design, and static-test data acquisition
- Directed design, analysis, and manufacturing of the first undergraduate axial turbojet through research, FEA, and CFD, targeting 80,000 RPM and 35 lbf of thrust

Engineers Without Borders, UPenn · Robotics Manufacturing Lead

Jan 2024 – Mar 2026

- Engineering a robotic system to safely collect broken glass in hazardous areas of Philadelphia
- Contributed to a sanitation system design for a Guatemalan school serving 90 students

Stirling Engine · MEAM 2010

May 2024

- Designed full engine assembly in SolidWorks with manufacturing plans from GD&T tolerance stackup; machined and assembled 18 parts in aluminum, brass, stainless steel, and walnut

EDUCATION

University of Pennsylvania

BSE + MSE, Mechanical Engineering

Aug 2023 – May 2027
Philadelphia, PA

GPA: 3.92

Minors in Mathematics & History

Coursework

Compressible Flow & Combustion · Aerodynamics · Turbulence · Heat & Mass Transfer · Fluid Mechanics · Mechanics of Solids · Machine Design & Manufacturing · FEA · Thermodynamics · Dynamics · Advanced Math

TEACHING

Teaching Assistant, IPD 5010

Graduate machine design & manufacturing

Aug 2025 – Present

Machine Lab Assistant

Precision Machining Laboratory: shop maintenance, design and CAM guidance for students

Aug 2024 – Present

SKILLS

Software

Siemens NX · SolidWorks (CSWA) · Onshape · ANSYS (CFD & FEA) · MATLAB · Python · Java (prior)

Manufacturing

ASME GD&T · CNC (MasterCAM, PT4) · 3-Axis Mill · Lathe · Injection Molding · Additive Manufacturing

ACTIVITIES

Wharton Undergrad Aerospace Club (WUAC)

Jan 2024 – Dec 2025

American Society of Mechanical Engineers (ASME)

Jan 2024 – Present

INTERESTS

Backpacking (30 countries) · Scuba Diving (PADI Open Water) · Karate (Sensei) · Baseball · Piano · World History